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Task Grouping Analysis

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1 Executive Summary

A task scheduling system could place similar DUR tasks in succession; reducing efficiency loss due to context switching.

Using historical RxBuddy DUR tasks from 2026-02-01 to 2026-08-01, it is estimated that the duration of a DUR task can be 3.4% less if it follows a DUR of the same cohort immediately than in the case it is preceded by a DUR of different cohort. See section 2.3 for more details.

The annualized potential savings in pharmacist labor value is estimated to be of the range $80,000–$88,000 per year. See section 2.4 obtained by applying the 3.4% reduction to successful DUR tasks that are not already preceded by DURs of the same cohort assuming maximum scheduling flexibility.

We note that the results are based on an observational study and not a randomized controlled experiment where the effects of confounding factors could be randomized away. In order to get a more definitive idea of possible improvement, a randomized controlled experiment is advised.

2 Appendix

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2.1 Reproducing the Numerical Results

The code to reproduce all the results is in [4]. Follow the instructions in the HOWTO.md file to generate the figures, models and analyses. You will need to clone the repository to run the code. The code uses make and python. In addition you need to have access to a Snowflake role that has access to the following tables:

```
EDLDB.PET_HEALTH_ANALYTICS_SANDBOX.FCT__TASKS_LIFECYCLE
EDLDB.BT_HCA_HCDM.RXP_RX_USAGE_ACTIONS
```

`CHEWYBI.CUSTOMER_PETPROFILES`
`EDLDB.PDM.PRODUCT`
`EDLDB.BT_HCA_HCDM.RXP_PRESCRIPTIONS`
`EDLDB.PET_HEALTH_ANALYTICS_SANDBOX.FCT__RX_RESULT_FIELD_CHANGE_SUMMARY`

and the pet health analytics sandbox in the dev environment;

`EDLDB_DEV.PET_HEALTH_ANALYTICS_SANDBOX`

to generate transient tables.

2.2 Data Preliminaries and Assumptions

The study considers DUR tasks recorded in `EDLDB.PET_HEALTH_ANALYTICS_SANDBOX.FCT__TASKS_LIFECYCLE` between 2026-02-01 and 2026-08-01.

Since the analysis is interested in figuring out the effect of a pharmacist repeatedly doing similar work, the study excludes DUR tasks that required a follow-up. We assume failed DURs break the concentration of the pharmacist and the effect of doing consecutive similar tasks. Table 1 shows the handoff-type distribution for all DUR tasks.

Only the actual time spent on the DUR is used (`DWELL_IN_PROGRESS_TO_CLOSED_SECONDS`). In the remainder of this report, that duration is called processing time or process duration.

Handoff type	DUR tasks	Processing time (seconds)	Hours
`NULL`	4,000,758	197,557,538	54,877.09
`PHARMACIST_FOLLOW_UP`	117,503	13,334,300	3,703.97
`TECHNICIAN_FOLLOW_UP`	6,736	820,191	227.83
`LEAD_FOLLOW_UP`	3,489	383,589	106.55
All DURs	**4,128,486**	**212,095,618**	**58,915.45**

Table 1: DUR task counts and imputed processing time by handoff type.

For analysis purposes we do not consider DUR tasks that happen simultaneously. Table 2 shows the intersection status of DUR tasks with no handoff type.

Intersection status	DUR tasks	Processing time (seconds)	Hours
Intersecting	332,599	16,046,580	4,457.38
Not intersecting	3,668,159	181,510,958	50,419.71
All DURs with NULL handoff type	**4,000,758**	**197,557,538**	**54,877.09**

Table 2: DUR task counts and imputed processing time by intersection status for tasks with no handoff type.

Finally, valid DURs at or above the approximate 95th-percentile processing-time cutoff of 156.608019 seconds are excluded as outliers. Table 3 shows the valid DURs on each side of the cutoff and reconciles them to all non-intersecting DURs with no handoff type. The other excluded group is calculated by subtraction; the aggregate does not separate its individual exclusion reasons.

Duration group	DUR tasks	Processing time (seconds)	Hours
Below cutoff	3,433,218	89,130,671	24,758.52
At or above cutoff	180,803	88,958,419	24,710.67
Other excluded (non-intersecting)	54,138	3,421,868	950.52
All non-intersecting DURs	**3,668,159**	**181,510,958**	**50,419.71**

Table 3: Reconciliation of non-intersecting DURs with no handoff type to valid DURs around the 156.608019-second cutoff.

2.3 Modeling DUR Processing Time

In this section, we develop a model that incorporates the following factors

- whether the task was preceded by a task of similar cohort
- pharmacist
- pet type
- if there were alterations done during the DUR process
- date effects
- approval and prescription channels
- cohort
- day of week

We find that being preceded by a task of the same cohort reduces DUR duration about 3.4% assuming everything else stays the same with the task.

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The model predicts that the more important factors are; cohort, pet type, whether alterations were needed, prescription source, the pharmacist, the date and approval channel. Day of the week is not found to be a significant factor.

We note that this is not a causal model nor is the data obtained from a randomized controlled experiment setup. Even though extreme care has been taken to reduce confounding, it may still be present.

2.3.1 Data

The model uses 3,433,218 DUR tasks with durations below the global 95th-percentile cutoff of 156.608019 seconds.

Let Y_i be duration in seconds, S_i the task's position in its same-cohort batch, and L_i the batch length. The sequence indicators are

$$P_i = \mathbf{1}(L_i > 1 \wedge S_i > 1), \qquad F_i = \mathbf{1}(L_i > 1 \wedge S_i < L_i).$$

`SAME_PRECEDING` (P_i) means an earlier task in the same-cohort batch precedes task i. `SAME_FOLLOWING` (F_i) means a later task in that batch follows it. Consecutive tasks for a user must have no break longer than one hour; overlapping and failed DUR tasks are excluded. Both indicators can equal one. Missing parent part numbers form a separate category.

Indicator	Value	Tasks	Share of sample
SAME_PRECEDING	0	1,875,931	54.64%
SAME_PRECEDING	1	1,557,287	45.36%
SAME_FOLLOWING	0	1,878,842	54.73%
SAME_FOLLOWING	1	1,554,376	45.27%

2.3.2 Model

The outcome is the natural logarithm of duration:

$$\log(Y_i) = \alpha + \beta_P P_i + \beta_F F_i + \beta_C C_i + \gamma_{c_i} + \delta_{p_i} + \theta_{w_i} + \phi_{a_i} + \psi_{s_i} + b_{u(i)} + h_{d(i)} + q_{r(i)} + \epsilon_i.$$

- Y_i: positive imputed DUR duration, in seconds, for task i.
- α: global intercept for the reference categories, zero indicators, and zero random intercepts.
- P_i, F_i: same-preceding and same-following indicators above; β_P, β_F are their conditional log-duration coefficients.
- $C_i = \mathbf{1}(N_i > 0)$: whether task i has at least one correction; β_C is its coefficient.
- $\gamma_{c_i}, \delta_{p_i}, \theta_{w_i}, \phi_{a_i}, \psi_{s_i}$: additive fixed effects for cohort, pet type, weekday, approval channel, and prescription source. Their references are Cohort 1, Dog, Sunday, `PH`, and `PRACTICE_HUB`. There are no fixed-effect cells or interactions.
- $b_{u(i)}, h_{d(i)}, q_{r(i)}$: crossed random intercepts for user, process date, and parent part number. They are independent normals with mean zero and variances $\sigma_u^2, \sigma_d^2, \sigma_r^2$.
- ϵ_i: independent normal task residual with mean zero and common variance σ_e^2.

2.3.3 Numerical procedure

The global approximate 95th-percentile cutoff is applied to all otherwise eligible positive durations; equality with the cutoff is excluded. Retained durations are log-transformed. Categorical variables are encoded as contrasts to the stated references and the three grouping variables as crossed random intercepts.

Restricted maximum likelihood (REML) estimates fixed coefficients and variance components. The implementation accumulates cross-products and profiles out fixed coefficients and residual variance at each candidate set of random-effect variance ratios. Cholesky solves evaluate the profiled objective; bounded L-BFGS-B minimizes it on the log-variance-ratio scale. The fit converged.

Fixed-effect standard errors come from the fitted covariance matrix. The two-sided p-values and 95% confidence intervals use normal Wald approximations, with $z = \hat{\beta}/\mathrm{SE}(\hat{\beta})$. A coefficient is marked significant when its interval excludes zero. Random-intercept SD intervals use observed information on the log-SD scale; the residual-SD interval uses a chi-square approximation. All estimates and SDs are on the log scale.

2.3.4 Fixed-effect coefficients

`AC` means approval channel relative to `PH`; `PS` means prescription source relative to `PRACTICE_HUB`. Cohort and pet-type contrasts use Cohort 1 and Dog as references. Yellow highlights significant rows.

Term	Estimate	SE	p-value	95% CI	Significant
SAME_PRECEDING	-0.034871	0.000858	$< 10^{-300}$	[-0.036552, -0.033189]	Yes
SAME_FOLLOWING	-0.001013	0.000858	0.237696	[-0.002695, 0.000669]	No
HAS_CORRECTION	0.335919	0.001002	$< 10^{-300}$	[0.333956, 0.337883]	Yes
WEEKDAY_MON	0.007606	0.024733	0.758432	[-0.040869, 0.056081]	No
WEEKDAY_TUE	0.018008	0.024728	0.46647	[-0.030459, 0.066475]	No
WEEKDAY_WED	0.013373	0.024734	0.588742	[-0.035105, 0.061851]	No
WEEKDAY_THU	0.010322	0.024736	0.67648	[-0.038161, 0.058804]	No
WEEKDAY_FRI	0.008029	0.024735	0.745478	[-0.040450, 0.056509]	No

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Term	Estimate	SE	p-value	95% CI	Significant
WEEKDAY_SAT	0.003198	0.024754	0.897221	[-0.045320, 0.051715]	No
Coh: Cohort 2 vs Cohort 1	0.208007	0.016863	5.85261×10^{-35}	[0.174956, 0.241058]	Yes
Coh: Cohort 3 vs Cohort 1	0.504763	0.017907	8.19508×10^{-175}	[0.469666, 0.539860]	Yes
Coh: Cohort 4 vs Cohort 1	0.548922	0.035367	2.50295×10^{-54}	[0.479605, 0.618240]	Yes
Coh: Unknown vs Cohort 1	0.610945	0.034303	5.86458×10^{-71}	[0.543713, 0.678177]	Yes
Pet: Missing vs Dog	-0.809849	0.542986	0.135837	[-1.874083, 0.254384]	No
Pet: Bird vs Dog	0.430595	0.032426	3.05603×10^{-40}	[0.367041, 0.494149]	Yes
Pet: Cat vs Dog	0.053896	0.002680	5.71345×10^{-90}	[0.048644, 0.059148]	Yes
Pet: Farm Animal vs Dog	0.508492	0.192162	0.008141	[0.131862, 0.885123]	Yes
Pet: Fish vs Dog	0.566626	0.146524	0.000110128	[0.279444, 0.853808]	Yes
Pet: Horse vs Dog	-0.023698	0.010410	0.0228128	[-0.044101, -0.003296]	Yes
Pet: Reptile vs Dog	0.314481	0.050987	6.92146×10^{-10}	[0.214549, 0.414414]	Yes
Pet: Small Pet vs Dog	0.322064	0.012358	1.02986×10^{-149}	[0.297842, 0.346287]	Yes
AC: CLINIC_EMAIL vs PH	0.299482	0.057477	1.88374×10^{-7}	[0.186829, 0.412134]	Yes
AC: CUSTOMER_MAIL vs PH	0.600193	0.016604	4.01198×10^{-286}	[0.567650, 0.632735]	Yes
AC: DIGITAL vs PH	0.190563	0.051772	0.000232496	[0.089092, 0.292034]	Yes
AC: FAX vs PH	0.258302	0.014273	3.33691×10^{-73}	[0.230328, 0.286277]	Yes
AC: PARTNER vs PH	-0.043703	0.010128	1.59659×10^{-5}	[-0.063555, -0.023852]	Yes
AC: RHAPSODY vs PH	0.094970	0.066836	0.155337	[-0.036027, 0.225966]	No
AC: VERBAL vs PH	-0.020313	0.013558	0.134062	[-0.046885, 0.006259]	No
AC: VET vs PH	0.012257	0.134401	0.927334	[-0.251165, 0.275679]	No
PS: CLINIC_EMAIL vs PRACTICE_HUB	-0.004722	0.014214	0.739731	[-0.032581, 0.023137]	No
PS: CS_FIRST_CONTACT vs PRACTICE_HUB	-0.109942	0.259207	0.671457	[-0.617979, 0.398094]	No
PS: CUSTOMER_MAIL vs PRACTICE_HUB	0.054217	0.016288	0.000872503	[0.022294, 0.086141]	Yes
PS: CVC vs PRACTICE_HUB	0.027068	0.066757	0.685129	[-0.103774, 0.157911]	No
PS: DIGITAL vs PRACTICE_HUB	0.025127	0.050985	0.622127	[-0.074801, 0.125055]	No
PS: FAX vs PRACTICE_HUB	-0.007227	0.014269	0.612511	[-0.035193, 0.020739]	No
PS: PARTNER_API vs PRACTICE_HUB	0.026072	0.010098	0.00982729	[0.006280, 0.045863]	Yes
PS: RHAPSODY vs PRACTICE_HUB	-0.039594	0.079752	0.619565	[-0.195905, 0.116717]	No
PS: TELEMEDICINE vs PRACTICE_HUB	0.236886	0.067306	0.000432265	[0.104970, 0.368803]	Yes
PS: VERBAL vs PRACTICE_HUB	-0.004447	0.014919	0.76563	[-0.033689, 0.024794]	No

2.3.5 Random-effects statistics

Component	SD (log scale)	Approximate 95% CI
User random intercept	0.545034	[0.503849, 0.589586]
Date random intercept	0.088873	[0.079975, 0.098762]
Parent-part-number random intercept	0.169522	[0.160562, 0.178982]
Residual	0.767615	[0.767041, 0.768189]

2.3.6 Fit statistics

The global intercept is $\hat{\alpha} = 3.034533$ log seconds. It applies to the reference categories, no correction, neither sequence indicator, and zero random intercepts.

Quantity	Value
Tasks	3,433,218
Users	340
Dates	182
Parent part numbers	885
Conditional R^2 (log scale)	0.300786
Conditional RMSE (log seconds)	0.767469
REML negative log-likelihood	3,966,831.492274
Converged	Yes

The conditional fit measures include estimated random intercepts.

2.3.7 Interpretation and assumptions

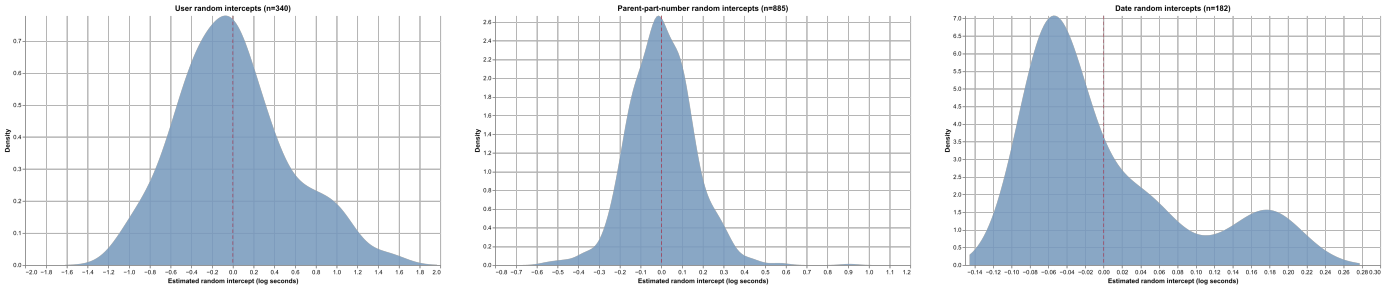
Exponentiating a sequence coefficient gives its conditional duration ratio: $\exp(\hat{\beta}_P) = 0.965730$. The model assumes additive effects, independent normally distributed random intercepts and residuals, and common residual variance.

2.3.8 Random-intercept distributions

Figure 1 shows the fitted intercept distributions for users (a), parent part numbers (b), and dates (c). Each grouping level has equal weight within its panel; the dashed vertical line marks zero.

The users (pharmacists) and dates random effects suggest bimodal distributions and invite further analysis.

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(a) User **(b) Parent part number** **(c) Date**

Figure 1: Fitted random-intercept densities on the log-duration scale.

2.4 Estimating the Potential Cost Savings of Grouping Similar Tasks

In this section, we apply the 3.4% reduction to a subset of DUR tasks to arrive at a labor savings in the range $80,000–$88,000 per year. See section 2.4.3 for more details.

The labor savings not only depend on the 3.4% rate of reduction but also on the DUR tasks that are eligible to benefit from this via minimum context switching.

In the first following section, we discuss the selection of eligible tasks in detail.

After that, we discuss the assumptions on scheduling and the logic to compute the potential labor savings on the eligible population.

We finish with the presentation of the results.

2.4.1 Determining the Eligible DUR Tasks For Duration Reduction

In this section we discuss, the historical DUR tasks that were deemed candidates for the hypothetical duration reduction.

Figures 2 and 3 summarize DUR task counts and imputed processing time. DURs that required a hand-off were excluded assuming that the pharmacist would lose concentration in this case. The “Possible multitasking” group was excluded because the pharmacist was working on multiple DURs or tasks at the same time, and we assume that any context switching benefit would be lost.

DURs with processing times more than 156.6 seconds were excluded. These DURs have durations above the 95th percentile threshold and deemed abnormal. The pharmacist is possibly multitasking.

The `SAME_PRECEDING=1` denotes historical DUR tasks that may already benefit from following a DUR of the same cohort.

The `SAME_PRECEDING=0` tasks are subjected to the 3.4% reduction in DUR times.

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The leading number on each region is its identifier in both figures:

1	All DURs	7	Singletasking
2	No handoff	8	Other validity exclusions
3	Pharmacist follow-up	9	At or above cutoff
4	Technician follow-up	10	Below cutoff
5	Lead follow-up	11	`SAME_PRECEDING = 1`
6	Possible multitasking	12	`SAME_PRECEDING = 0`

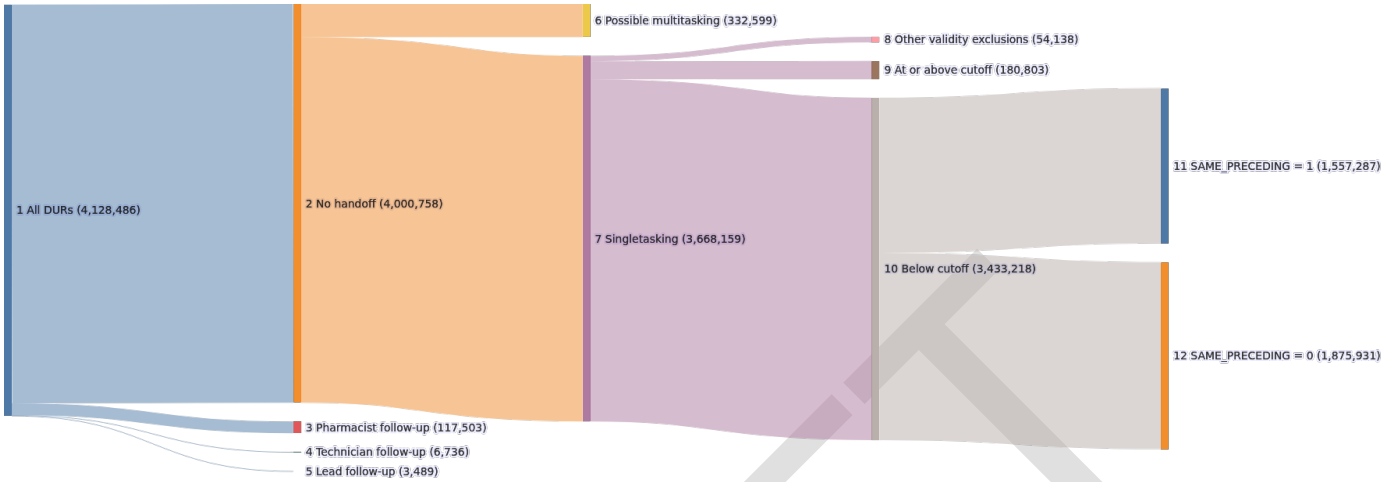


Figure 2: DUR task counts partitioned by handoff type, possible multitasking or singletasking, validity, the 156.608019-second cutoff, and `SAME_PRECEDING`.

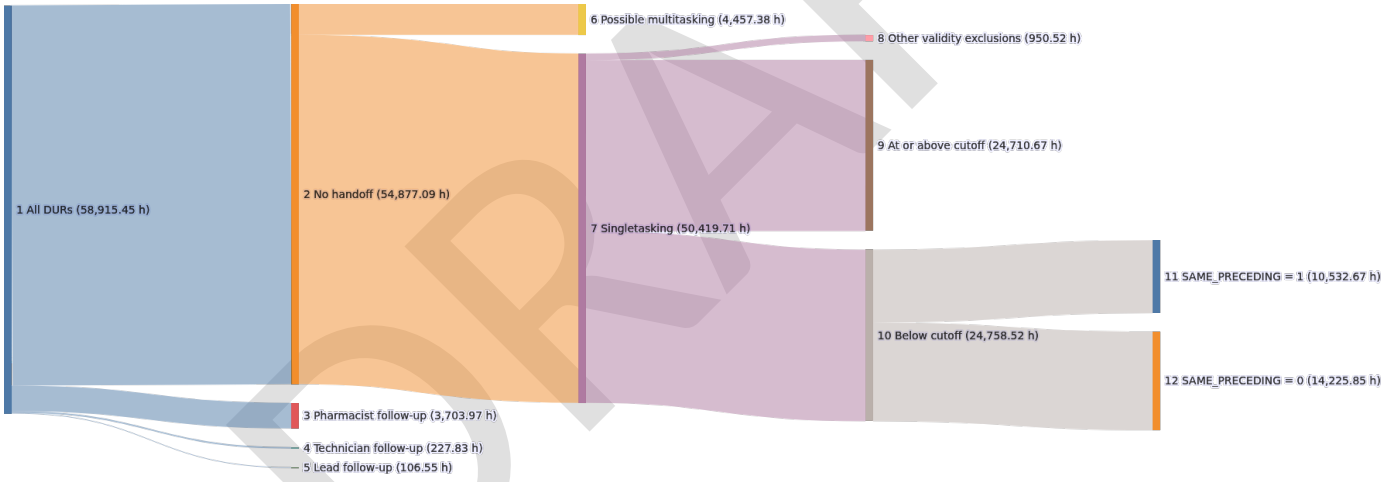


Figure 3: DUR processing time partitioned by the same stages. Flow widths use imputed processing seconds; node labels show hours rounded to two decimal places.

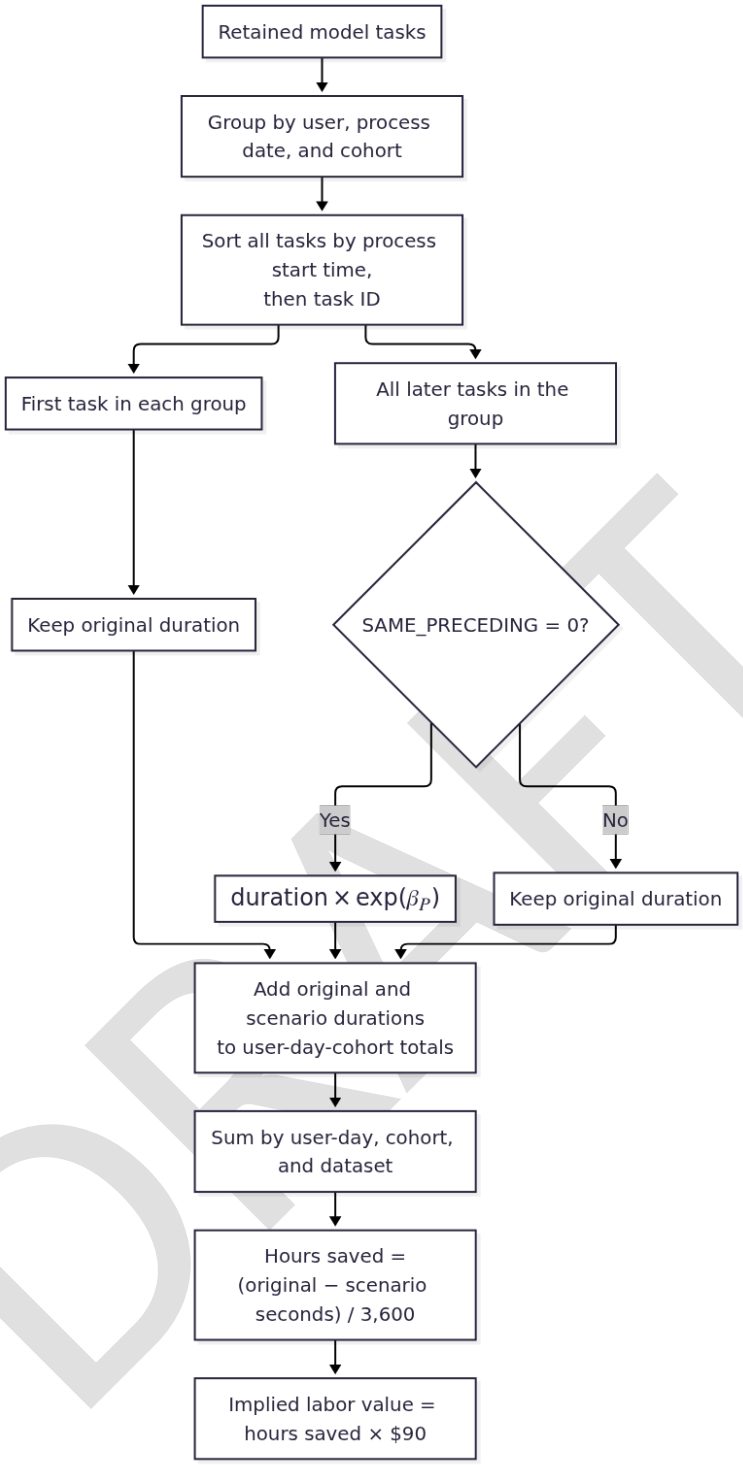
2.4.2 Reduction Methodology

In this section, we assume

- that the DUR tasks of a pharmacist can be rearranged for a given day.
- that stretches of successful DURs are not interrupted by multitasking or a DUR failure.

Figure 4 shows the calculation: within each user, process date, and cohort, the first task keeps its observed duration and later tasks with `SAME_PRECEDING = 0` receive the reduced duration obtained from the model.

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Figure 4: Same-preceding counterfactual algorithm. Tasks are ordered within each user-day-cohort group; the first task and tasks already marked `SAME_PRECEDING = 1` retain their observed durations.

2.4.3 Results

Tables 5 and 6 give the resulting totals for retained process dates from 2026-02-01 through 2026-08-01. The fitted duration ratio applied to eligible later tasks is $\exp(\beta_P) = 0.965730$. See section 2.3 for how this value was computed.

Quantity	Value
Retained DUR tasks	3,433,218
User-day-cohort groups / unchanged first tasks	64,681
Later `SAME_PRECEDING = 0` tasks adjusted	1,814,117
Original processing time (seconds)	89,130,671.0
Scenario processing time (seconds)	87,458,136.9
Time reduction (hours)	464.6
Change from original	-1.88%
Pharmacist cost per hour	$90.00
Implied labor value	$41,813.35

Table 5: Overall same-preceding counterfactual for the retained model sample.

Cohort	Tasks	Adjusted	Original (s)	Scenario (s)	Saved (h)	Value ($)
Cohort 1	1,904,176	862,667	43,800,431.0	43,103,502.2	193.6	17,423.22
Cohort 2	1,331,432	791,247	37,428,756.0	36,667,828.8	211.4	19,023.18
Cohort 3	159,018	132,205	6,121,581.0	5,950,003.0	47.7	4,289.45
Cohort 4	17,615	13,962	868,270.0	845,453.7	6.3	570.41
Unknown	20,977	14,036	911,633.0	891,349.3	5.6	507.09
Total	3,433,218	1,814,117	89,130,671.0	87,458,136.9	464.6	41,813.35

Table 6: Counterfactual processing time and implied labor value by cohort, at $90 per pharmacist hour. Rounded cohort values may differ slightly from the total.

Table 7 varies only the `SAME_PRECEDING` coefficient over its fitted 95% confidence interval. Each row uses the same 1,814,117 adjusted tasks and $90 hourly rate. The source extract includes tasks created from 2026-02-01 inclusive through 2026-08-01 exclusive, a 181-day window. Annualized values multiply the period values by $365/181$, assuming the same average daily volume and task mix throughout the year.

Coefficient scenario	β_P	181-day value	365-day value
Lower 95% bound	-0.036552403	$43,793.22	$88,312.30
Fitted estimate	-0.034870727	$41,813.35	$84,319.74
Upper 95% bound	-0.033189052	$39,830.15	$80,320.46

Table 7: Period and annualized implied labor value under the fitted `SAME_PRECEDING` coefficient and its 95% confidence bounds, at $90 per pharmacist hour. Annualized values use unrounded period estimates.

2.5 Example Sequences of Similar DUR Tasks

In this section, two task sequences from the same user are examined.

Both sequences consider prescriptions with Parasiticide as merchandising category level 3 (MC3) and with approval channel as Practice Hub. All prescriptions belong to cohort 1. The `Updated` field column contains the data the pharmacist altered during the DUR.

In the first sequence, all the prescriptions are for dogs. (Table 8). If the pharmacist updates prescription information, the DUR process highly likely takes longer. The 13 DUR long task sequence takes place within 7 minutes and the DUR durations vary between 6 and 97 seconds suggesting that there is significant variation. It is hard to discern any pattern that suggests any benefit of grouping tasks together.

In the second sequence, the prescriptions are for cats (Table 9). Note that for the same MC3 and cohort (cohort 1 in this case), pet and even part number (158995) we can expect significant variation. In this case, it can be seen that the duration varies between 3 and 50 seconds for the same part number and pet type and the same pharmacist in a 4 minute time span.

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Order	Part no.	DUR (s)	Start (UTC)	Updated
1	146066	40	2026-04-30 06:42:54	
2	152702	27	2026-04-30 06:43:36	directions
3	146391	40	2026-04-30 06:44:06	
4	151583	7	2026-04-30 06:44:48	
5	146138	7	2026-04-30 06:44:56	
6	146142	43	2026-04-30 06:45:05	directions
7	146142	35	2026-04-30 06:45:49	directions
8	146064	39	2026-04-30 06:46:26	directions
9	146167	12	2026-04-30 06:47:07	
10	146322	9	2026-04-30 06:47:21	directions
11	146167	97	2026-04-30 06:47:32	directions
12	146050	6	2026-04-30 06:49:11	
13	146387	54	2026-04-30 06:49:19	

Table 8: Sequence A: dog prescriptions in cohort 1.

Order	Part no.	DUR (s)	Start (UTC)	Updated
1	158995	11	2026-07-15 02:17:23	
2	158995	35	2026-07-15 02:17:36	
3	158995	3	2026-07-15 02:18:13	
4	158995	13	2026-07-15 02:18:17	
5	158995	13	2026-07-15 02:18:31	
6	158995	6	2026-07-15 02:18:46	
7	158995	16	2026-07-15 02:18:53	
8	158995	13	2026-07-15 02:19:11	
9	158995	6	2026-07-15 02:19:25	
10	158995	50	2026-07-15 02:19:32	
11	158998	11	2026-07-15 02:20:23	
12	158998	29	2026-07-15 02:20:36	

Table 9: Sequence B: cat prescriptions in cohort 1.

2.6 Cohort Definitions

This section is verbatim reproduced from the conversation with JD Wright to provide background on the history of cohorts and their rationale.

Cohort 1: Boxed parasiticides. This was the best starting point because it represents roughly 40% of total DUR volume and the majority of the initial digital volume, while the clinical rules are relatively standardized and label-driven. The primary checks are species, weight range, monthly directions, days' supply, duplicate parasite coverage, and known contraindications. This gave us a high-volume population with relatively low variability for validating the core RxBuddy and DUR Copilot experience.

Cohort 2: Non-controlled, non-boxed medications. This represents roughly 29% of total DUR volume and materially expands coverage, but the clinical review is more complex. These medications require generalized dosing evaluation, including strength, dose, frequency, route, indication, species appropriateness, duplicate therapy, contraindications, and potential interactions. We separated this cohort so those broader AI checks could be validated without also introducing compounding or controlled-substance requirements at the same time.

Cohort 3: Non-controlled compounded medications. This is a smaller segment at roughly 4.4% of DUR volume, but it has distinct clinical and compliance requirements. Compounds require a documented compounding reason, formulation-specific directions, and validation that the prescribed days' supply does not exceed the product's beyond-use date. BUD rules vary by dosage form, storage conditions, formulation, and stability-study results. Keeping compounds separate allowed us to test these specialized workflows and manage their impact on pharmacist throughput independently.

Cohort 4: Gabapentin and the high-scrutiny regulatory workflow. This represents roughly 1.1% of DUR volume. The implemented Phase 4 scope focused on uncontrolled gabapentin rather than the full controlled-substance population, but it was separated because it introduces site eligibility, pharmacist skill gating, regulatory indicators, mixed-order handling, and state-specific scrutiny. It also gave us a contained way to validate infrastructure that will be needed for broader controlled-substance support later.

2.7 One-Sided Paired Comparison Test

In this section, a paired-test is carried out to determine if a task preceded by the same task is executed faster. The tests show that for cohort 1,2,3 there is a non-zero improvement but for cohort 4, the test cannot reject that there is no improvement.

The method does not try to test for confounders [3] which are possible factors affecting both the decision of scheduling one task after another and the output which is duration. In this study, we try to account for the user and the cohort, but we are not able to control for day of the week, pet or part number which might be in the set of confounders. Confounders can change the analysis result significantly. A similar approach in [1] points out that there is statistical reason to believe that controlling for cohort and user still may not eliminate the effect of confounders.

2.7.1 Purpose and method

This analysis compares Same Cohort and Different Cohort DUR work time within users, separately for Cohorts 1–4. Task durations are first averaged within each user, cohort, and allocation pattern. Only users with both allocation patterns in the same cohort are retained, so the comparison is paired by user.

The paired test is a one-sided paired t-test. The null hypothesis is that Same Cohort is not faster than Different Cohort (Same Cohort $-$ Different Cohort ≥ 0); the alternative is that this difference is less than zero, with $\alpha = 0.05$. The density plots in Figures 5–8 show the two distributions of user-level means on a base-10 logarithmic horizontal axis.

The 95% two-sided confidence intervals for each allocation-pattern mean and the paired mean difference use Student's t distribution and the matched users as the units of analysis. The one-sided 95% upper bounds correspond to the directional paired tests.

The source query keeps transitions on or after 2026-08-01, DUR tasks with final status `CLOSED`, non-null user and start time, work

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time between 0.01 and 5 minutes, and known item cohorts. Rows without a previous cohort are excluded.

2.7.2 User-level work-time densities

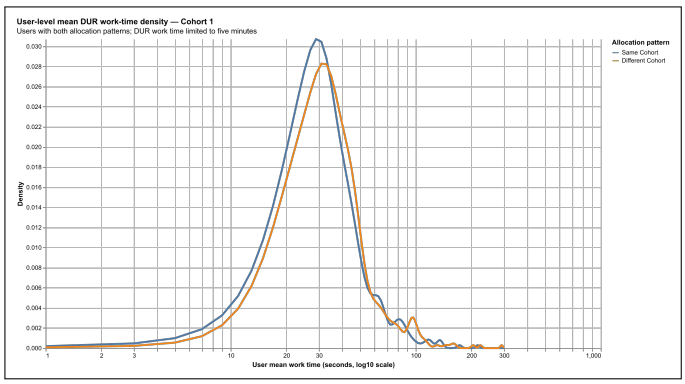


Figure 5: User-level work-time density, Cohort 1.

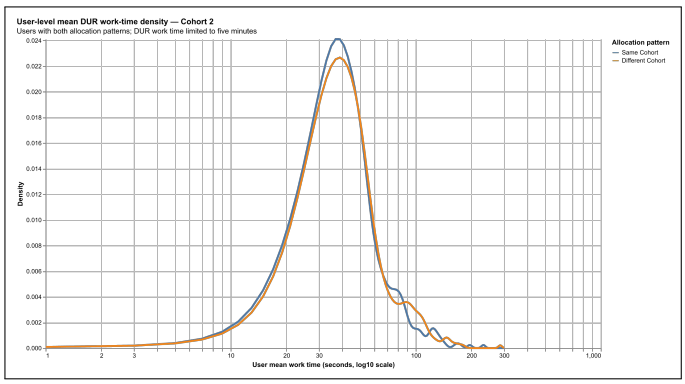


Figure 6: User-level work-time density, Cohort 2.

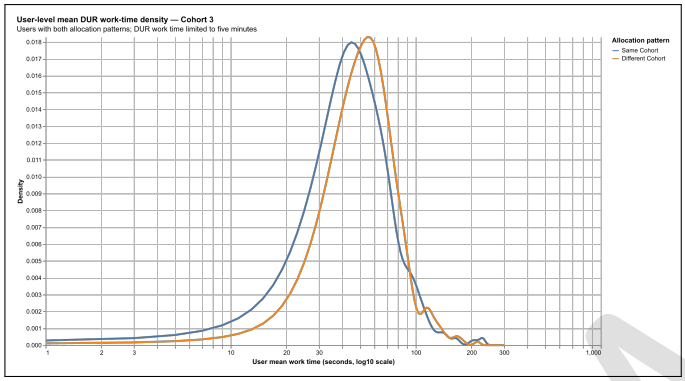


Figure 7: User-level work-time density, Cohort 3.

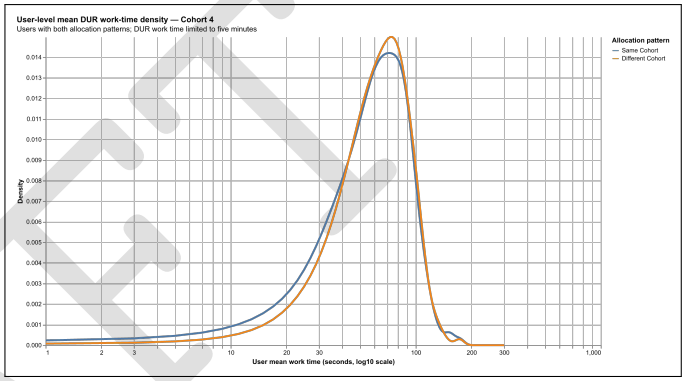


Figure 8: User-level work-time density, Cohort 4.

2.7.3 Cohort results

Table 10 reports the distribution of user-level mean work times. The same matched users contribute to both rows for each cohort.

Cohort	Allocation pattern	Users	Mean user mean (min)	95% CI for mean (min)	SD	Median	P25	P75
1	Same Cohort	289	0.6681	$[0.6193, 0.7170]$	0.4217	0.5410	0.4329	0.7469
1	Different Cohort	289	0.7308	$[0.6703, 0.7912]$	0.5220	0.5909	0.4648	0.7776
2	Same Cohort	286	0.8402	$[0.7831, 0.8974]$	0.4910	0.7114	0.5552	0.9312
2	Different Cohort	286	0.8767	$[0.8155, 0.9380]$	0.5259	0.7445	0.5660	0.9848
3	Same Cohort	250	1.0091	$[0.9399, 1.0783]$	0.5552	0.8809	0.6676	1.1693
3	Different Cohort	250	1.0650	$[1.0054, 1.1247]$	0.4785	0.9695	0.7510	1.2296
4	Same Cohort	139	1.2037	$[1.1285, 1.2790]$	0.4488	1.1814	0.9098	1.4688
4	Different Cohort	139	1.2202	$[1.1504, 1.2899]$	0.4161	1.2020	0.9106	1.4989

Table 10: User-level mean DUR work time by cohort and allocation pattern.

The paired comparison uses the within-user difference Same Cohort − Different Cohort. Table 11 gives the one-sided test results. At the 0.05 level, the null hypothesis is rejected for Cohorts 1–3 but not for Cohort 4.

Cohort	Users	Mean difference (min)	95% two-sided CI (min)	95% one-sided upper bound	t	df	One-sided p
1	289	-0.0626	$[-0.0972, -0.0281]$	$(-\infty, -0.0337]$	-3.5706	288	0.000208592
2	286	-0.0365	$[-0.0703, -0.0028]$	$(-\infty, -0.0082]$	-2.1302	285	0.0170078
3	250	-0.0559	$[-0.1043, -0.0075]$	$(-\infty, -0.0154]$	-2.2766	249	0.0118295
4	139	-0.0164	$[-0.0557, 0.0229]$	$(-\infty, 0.0165]$	-0.8272	138	0.204771

Table 11: One-sided paired tests of Same Cohort minus Different Cohort mean work time.

2.7.4 Data

The data is obtained using the following query.

```
WITH ordered AS (
  SELECT
    TASK_ID, USER_ID, ITEM_COHORT, STARTED_AT,
    TRANSITION_STARTED_AT::DATE AS dt,
```

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```
    LAG(ITEM_COHORT) OVER (
      PARTITION BY USER_ID, TRANSITION_STARTED_AT::DATE
      ORDER BY STARTED_AT
    ) AS prev_cohort,
    DWELL_IN_PROGRESS_TO_CLOSED_MINUTES AS work_min
  FROM EDLDB.PET_HEALTH_ANALYTICS_SANDBOX.FCT__TASKS_LIFECYCLE
  WHERE TRANSITION_STARTED_AT >= '2026-08-01'
    AND TASK_TYPE = 'DUR' AND FINAL_STATUS = 'CLOSED'
    AND USER_ID IS NOT NULL AND STARTED_AT IS NOT NULL
    AND DWELL_IN_PROGRESS_TO_CLOSED_MINUTES BETWEEN 0.01 AND 5
    AND ITEM_COHORT != 'Unknown'
)
SELECT
  TASK_ID AS task_id, USER_ID AS user_id, dt,
  CASE
    WHEN ITEM_COHORT = prev_cohort THEN 'Same Cohort'
    ELSE 'Different Cohort'
  END AS allocation_pattern,
  ITEM_COHORT AS cohort, work_min
FROM ordered
WHERE prev_cohort IS NOT NULL
ORDER BY allocation_pattern, work_min
```

2.8 Visual Summary of Factors Affecting DUR

This section contains a visual summary of the factors that affect DUR duration. The goal is to be exhaustive in detail knowing well that some of the factors cannot be measured. Figure 9 shows the hypothesized relationships. This graph guides the exploratory data analysis and the final model structure. The main factors affecting DUR duration are the pharmacist, the short and long term experience of the pharmacist (Pharmacist Past in the figure), the task itself containing; the prescription the pet, the product and other metadata, the software platform (Oracle or RxBuddy), time of day, time of the week, seasonality, management/process regulations and external disturbances such as inbound calls.

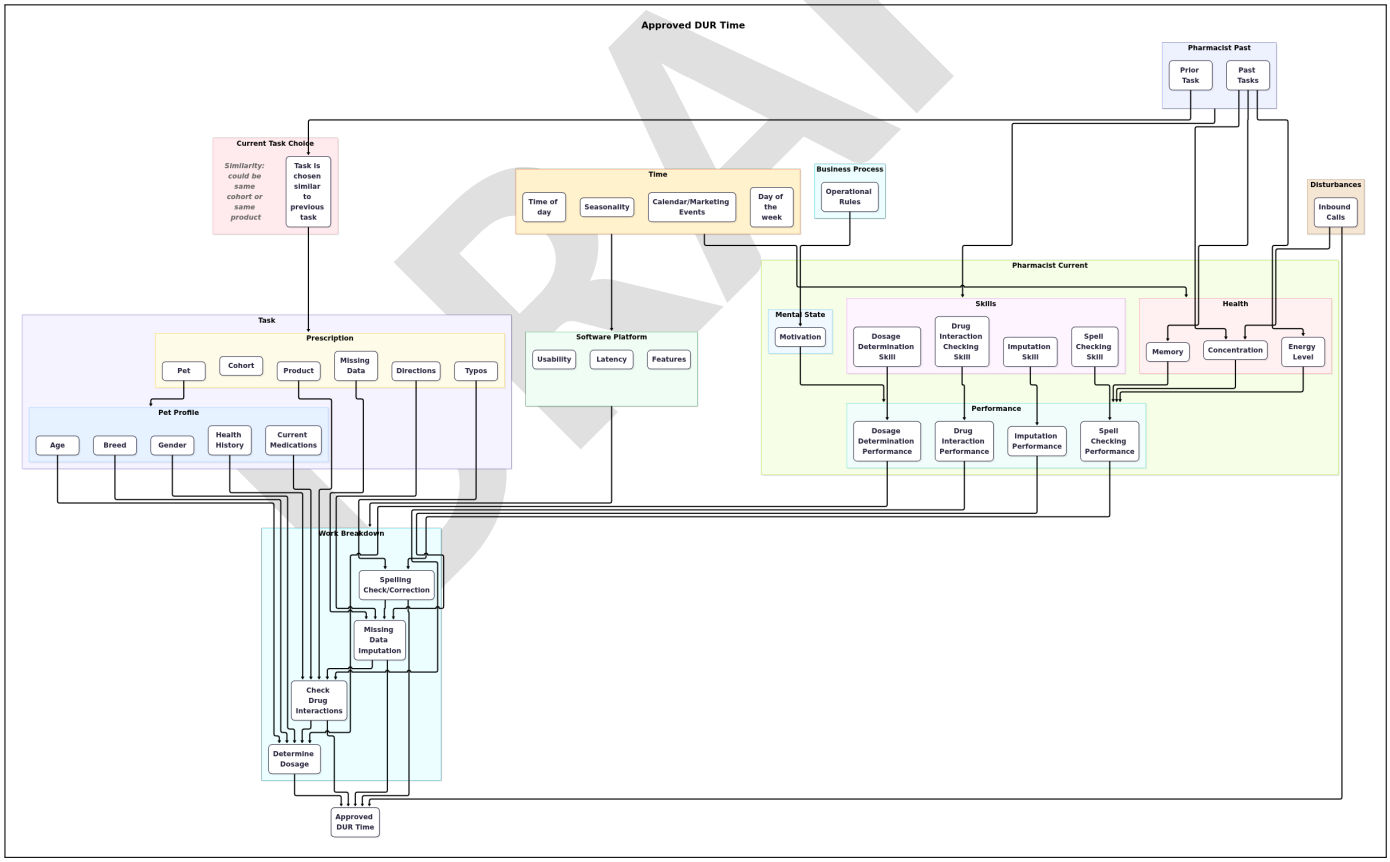


Figure 9: Hypothesized factors affecting approved DUR processing time.

2.9 Select Factors Affecting DUR Duration

We carried out extensive exploratory data analysis to understand the factors affecting DUR duration. The following sections contain the more prominent factors encountered. The causal graph section contains a visual summary of the hypothesized mechanisms that affect DUR duration.

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2.9.1 Pharmacist Performance

Pharmacists vary in their speed to process DURs (Figure 10). In addition, if a pharmacist is fast in handling a cohort 1 DUR, highly likely he/she is also fast handling other cohort DURs. Figure 11 shows the correlation of DUR median duration of 120 people who have carried out at least 30 tasks per cohort. As can be seen the performances correlate positively.

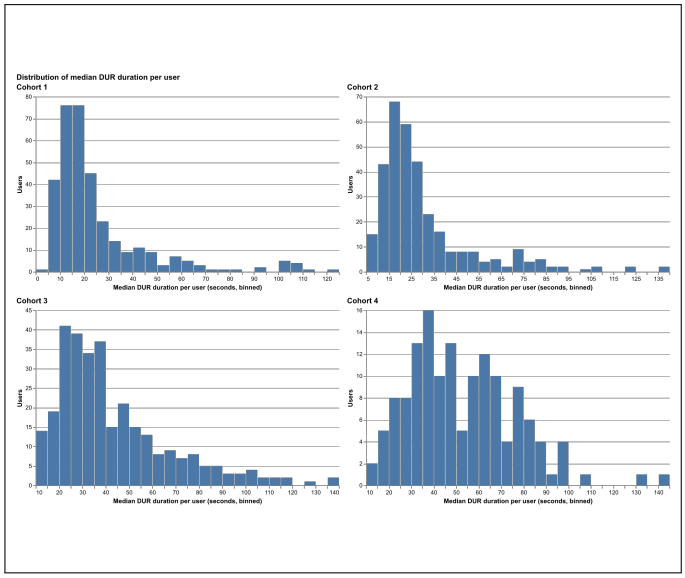


Figure 10: Distribution of per-pharmacist median DUR duration by cohort.

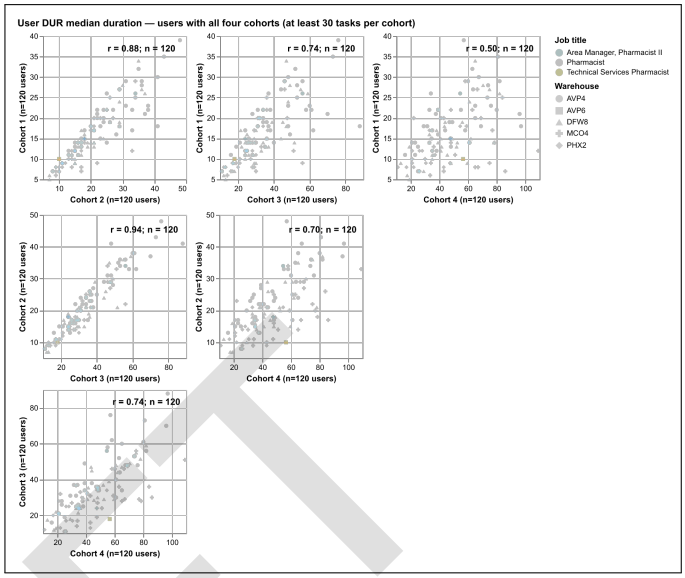
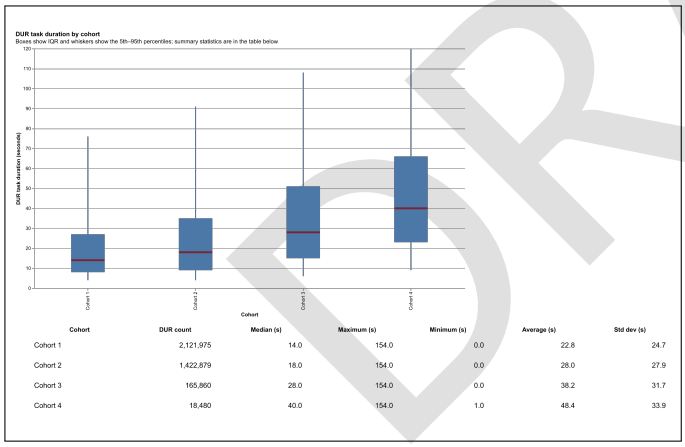


Figure 11: Median DUR duration across cohorts for 120 pharmacists.

2.9.2 Cohort and Part Number

Figure 12 compares task-level DUR durations across cohorts. Figure 13 compares part-number-level mean durations within each cohort. We can see that, within a cohort the DUR duration varies a lot. This is partly due to the pharmacist and the pet type.



Cohort	DUR count	Median (s)	Maximum (s)	Minimum (s)	Average (s)	Std dev (s)
Cohort 1	2,121,975	14.0	154.0	0.0	22.8	24.7
Cohort 2	1,422,879	18.0	154.0	0.0	28.0	27.9
Cohort 3	165,860	28.0	154.0	0.0	38.2	31.7
Cohort 4	18,480	40.0	154.0	1.0	48.4	33.9

Figure 12: Task-level DUR duration distributions by cohort.

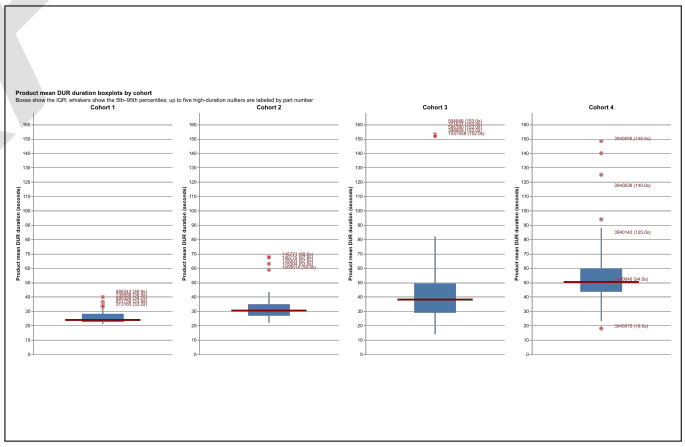
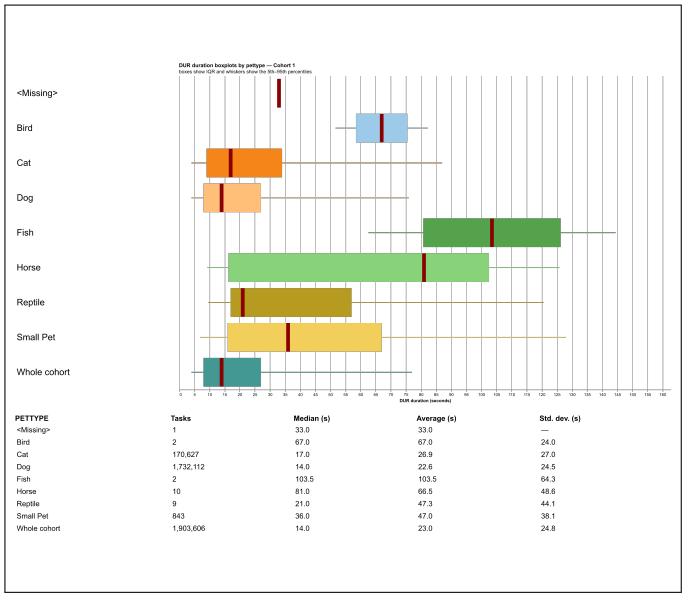


Figure 13: Part-number-level mean DUR duration distributions by cohort.

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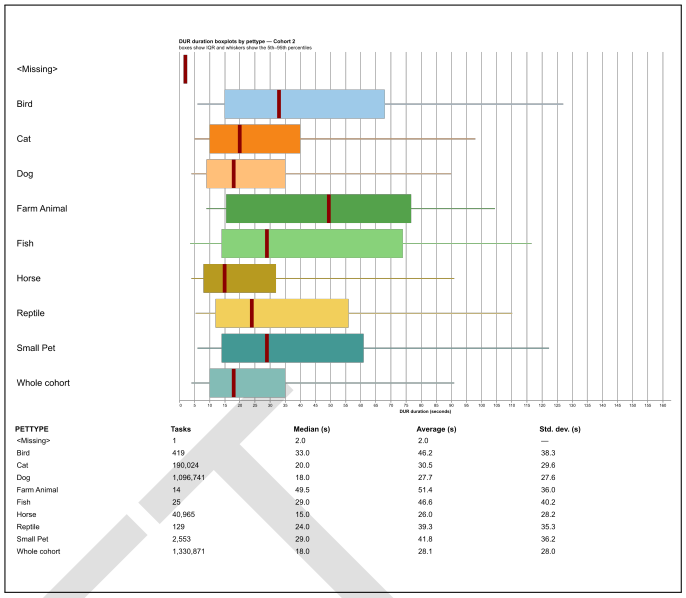
2.9.3 Pet Type

DUR durations vary across different pet types. Pharmacists seem to be able to process the more common dog, cat and horse related DURs faster on average as expected. The distributions for Cohorts 1–4 are shown in Figures 14–17, respectively.



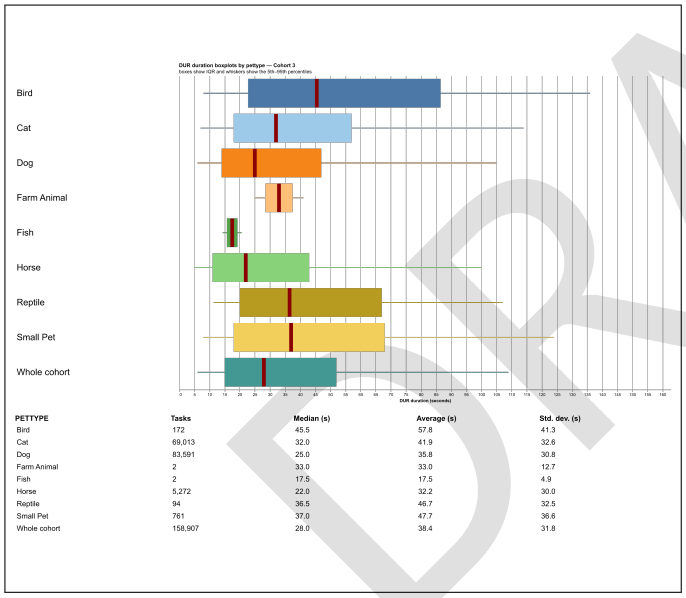
PETTYPE	Tasks	Median (s)	Average (s)	Std. dev. (s)
<Missing>	1	33.0	33.0	—
Bird	2	67.0	67.0	24.0
Cat	170,627	17.0	26.9	27.0
Dog	1,732,112	14.0	22.6	24.5
Fish	2	103.5	103.5	64.3
Horse	10	81.0	66.5	48.6
Reptile	9	21.0	47.3	44.1
Small Pet	843	36.0	47.0	38.1
Whole cohort	1,903,606	14.0	23.0	24.8

Figure 14: DUR duration by pet type, Cohort 1.



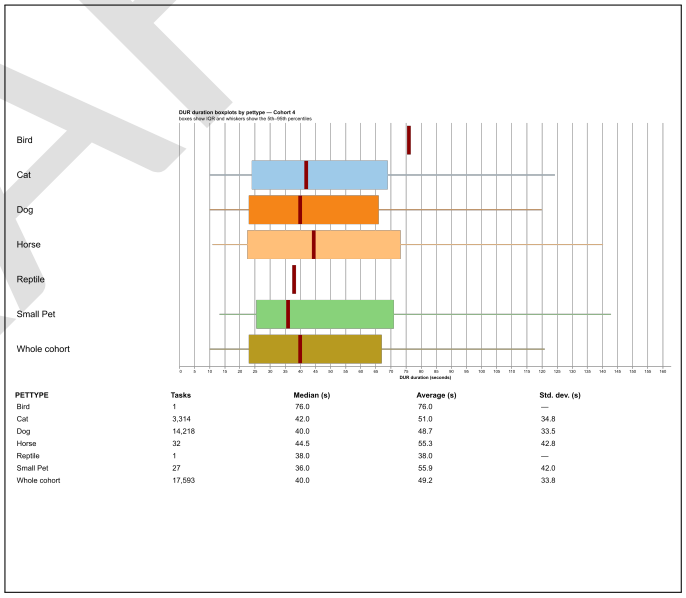
PETTYPE	Tasks	Median (s)	Average (s)	Std. dev. (s)
<Missing>	1	2.0	2.0	—
Bird	419	33.0	46.2	38.3
Cat	190,024	20.0	30.5	29.6
Dog	1,096,741	18.0	27.7	27.6
Farm Animal	14	49.5	51.4	36.0
Fish	25	29.0	46.6	40.2
Horse	40,965	15.0	26.0	28.2
Reptile	129	24.0	39.3	35.3
Small Pet	2,553	29.0	41.8	36.2
Whole cohort	1,330,871	18.0	28.1	28.0

Figure 15: DUR duration by pet type, Cohort 2.



PETTYPE	Tasks	Median (s)	Average (s)	Std. dev. (s)
Bird	172	45.5	57.8	41.3
Cat	69,013	32.0	41.9	32.6
Dog	83,591	25.0	35.8	30.8
Farm Animal	2	33.0	33.0	12.7
Fish	2	17.5	17.5	4.9
Horse	5,272	22.0	32.2	30.0
Reptile	94	36.5	46.7	32.5
Small Pet	761	37.0	47.7	36.6
Whole cohort	158,907	28.0	38.4	31.8

Figure 16: DUR duration by pet type, Cohort 3.



PETTYPE	Tasks	Median (s)	Average (s)	Std. dev. (s)
Bird	1	76.0	76.0	—
Cat	3,314	42.0	51.0	34.8
Dog	14,218	40.0	48.7	33.5
Horse	32	44.5	55.3	42.8
Reptile	1	38.0	38.0	—
Small Pet	27	36.0	55.9	42.0
Whole cohort	17,593	40.0	49.2	33.8

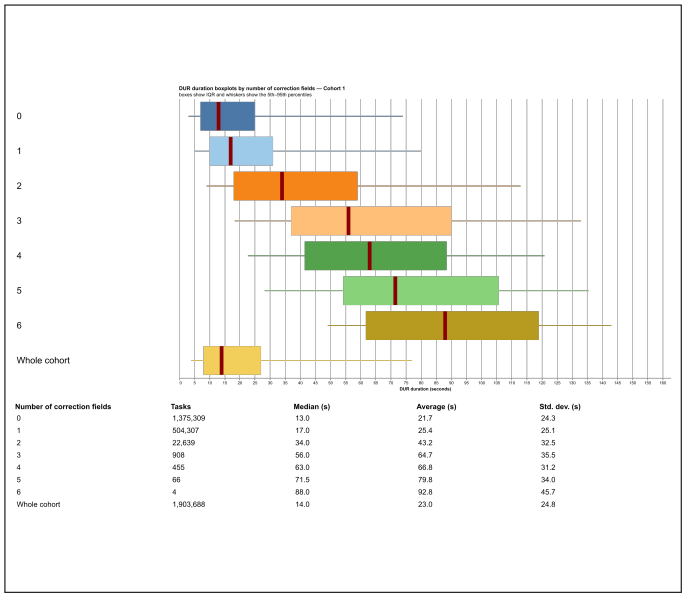
Figure 17: DUR duration by pet type, Cohort 4.

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2.9.4 Alterations/Corrections

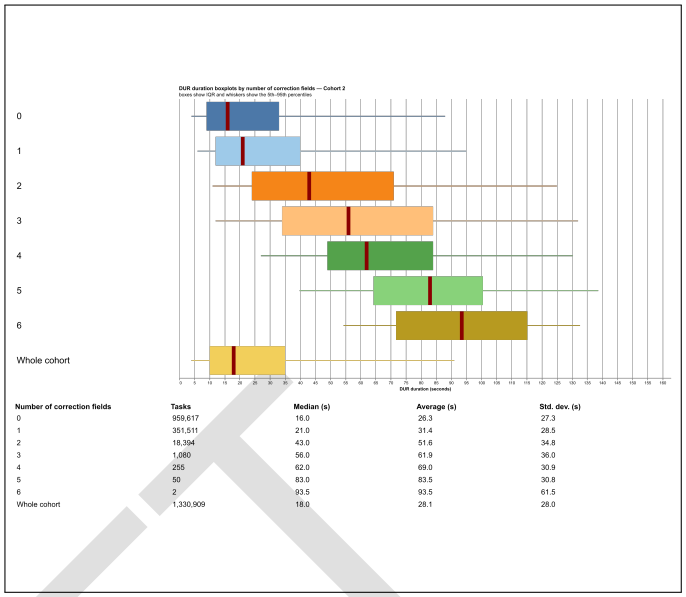
As expected, the more the pharmacist updates the prescription information, the longer the DUR process takes.

Figures 18–21 show DUR duration by the number of fields altered for Cohorts 1–4, respectively.



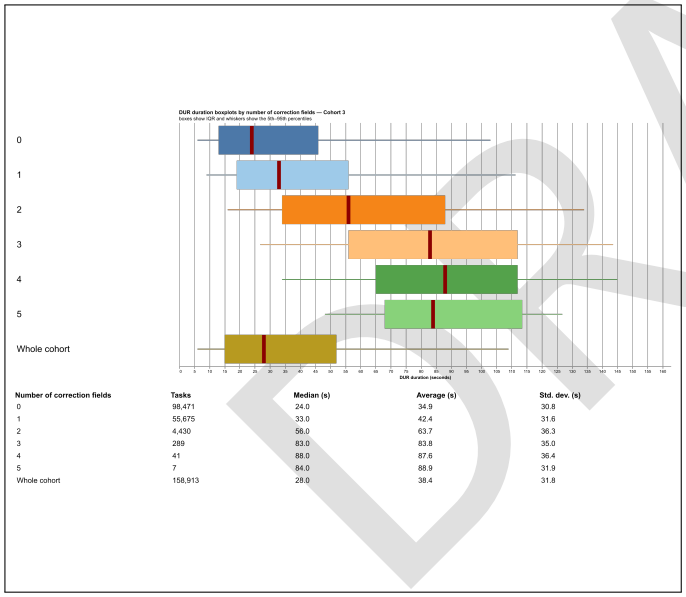
Number of correction fields	Tasks	Median (s)	Average (s)	Std. dev. (s)
0	1,375,309	13.0	21.7	24.3
1	504,307	17.0	25.4	25.1
2	22,639	34.0	43.2	32.5
3	908	56.0	64.7	35.5
4	455	63.0	66.8	31.2
5	66	71.5	79.8	34.0
6	4	88.0	92.8	45.7
Whole cohort	1,903,688	14.0	23.0	24.8

Figure 18: DUR duration by number of correction fields, Cohort 1.



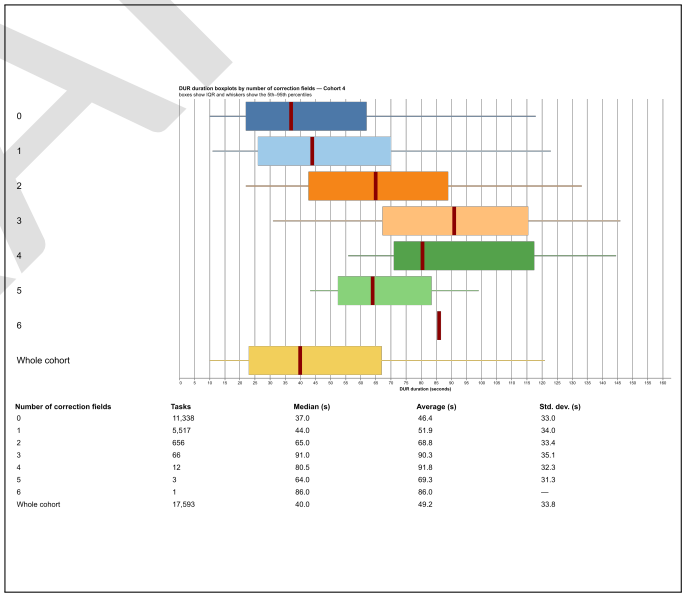
Number of correction fields	Tasks	Median (s)	Average (s)	Std. dev. (s)
0	959,617	16.0	26.3	27.3
1	351,511	21.0	31.4	28.5
2	18,394	43.0	51.6	34.8
3	1,080	56.0	61.9	36.0
4	255	62.0	69.0	30.9
5	50	83.0	83.5	30.8
6	2	93.5	93.5	61.5
Whole cohort	1,330,909	18.0	28.1	28.0

Figure 19: DUR duration by number of correction fields, Cohort 2.



Number of correction fields	Tasks	Median (s)	Average (s)	Std. dev. (s)
0	98,471	24.0	34.9	30.8
1	55,675	33.0	42.4	31.6
2	4,430	56.0	63.7	36.3
3	289	83.0	83.8	35.0
4	41	88.0	87.6	36.4
5	7	84.0	88.9	31.9
Whole cohort	158,913	28.0	38.4	31.8

Figure 20: DUR duration by number of correction fields, Cohort 3.

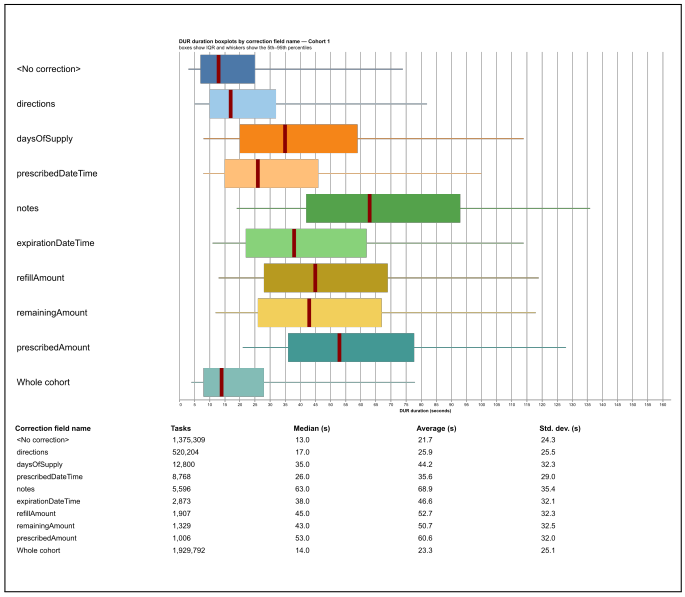


Number of correction fields	Tasks	Median (s)	Average (s)	Std. dev. (s)
0	11,338	37.0	46.4	33.0
1	5,517	44.0	51.9	34.0
2	656	65.0	68.8	33.4
3	66	91.0	90.3	35.1
4	12	80.5	91.8	32.3
5	3	64.0	69.3	31.3
6	1	86.0	86.0	—
Whole cohort	17,593	40.0	49.2	33.8

Figure 21: DUR duration by number of correction fields, Cohort 4.

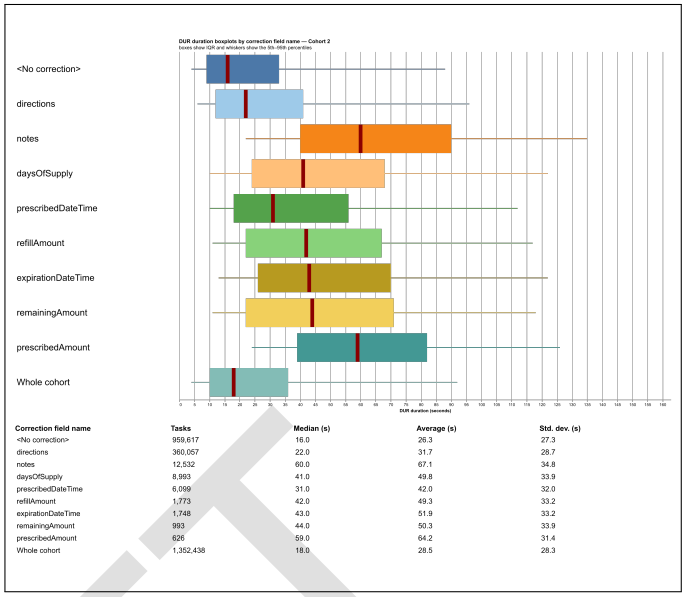
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Figures 22–25 show the corresponding distributions by the presence of that field name in the altered fields. This is not the contribution of each field by itself.



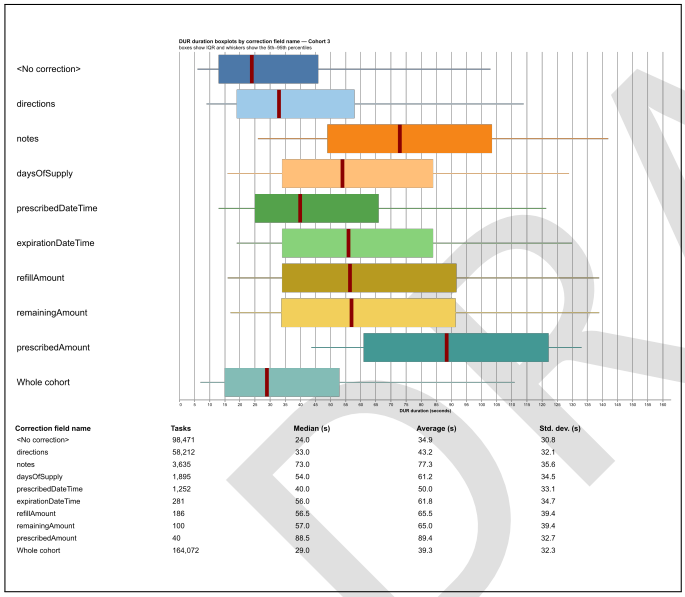
Correction field name	Tasks	Median (s)	Average (s)	Std. dev. (s)
<No correction>	1,375,309	13.0	21.7	24.3
directions	520,204	17.0	25.9	25.5
daysOfSupply	12,800	35.0	44.2	32.3
prescribedDateTime	8,768	26.0	35.6	29.0
notes	5,596	63.0	68.9	35.4
expirationDateTime	2,873	38.0	46.6	32.1
refillAmount	1,907	45.0	52.7	32.3
remainingAmount	1,329	43.0	50.7	32.5
prescribedAmount	1,006	53.0	60.6	32.0
Whole cohort	1,929,792	14.0	23.3	25.1

Figure 22: DUR duration by correction field name, Cohort 1.



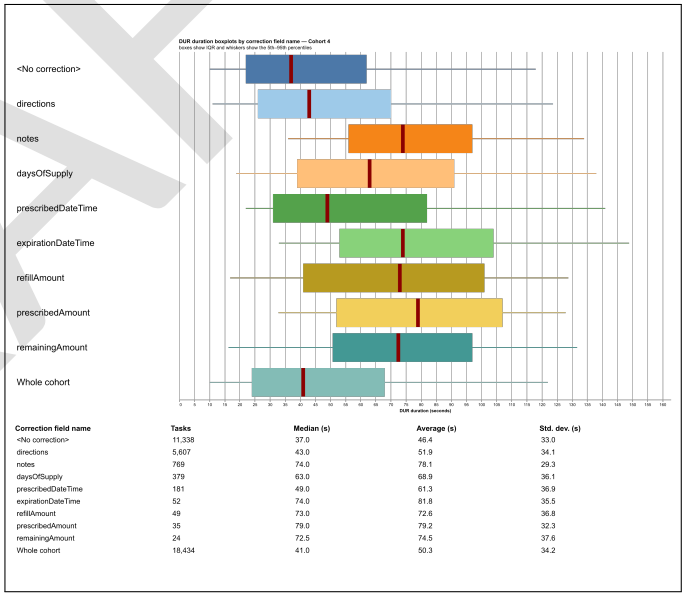
Correction field name	Tasks	Median (s)	Average (s)	Std. dev. (s)
<No correction>	959,617	16.0	26.3	27.3
directions	360,057	22.0	31.7	28.7
notes	12,532	60.0	67.1	34.8
daysOfSupply	8,993	41.0	49.8	33.9
prescribedDateTime	6,099	31.0	42.0	32.0
refillAmount	1,773	42.0	49.3	33.2
expirationDateTime	1,748	43.0	51.9	33.2
remainingAmount	993	44.0	50.3	33.9
prescribedAmount	626	59.0	64.2	31.4
Whole cohort	1,352,438	18.0	28.5	28.3

Figure 23: DUR duration by correction field name, Cohort 2.



Correction field name	Tasks	Median (s)	Average (s)	Std. dev. (s)
<No correction>	98,471	24.0	34.9	30.8
directions	58,212	33.0	43.2	32.1
notes	3,635	73.0	77.3	35.6
daysOfSupply	1,895	54.0	61.2	34.5
prescribedDateTime	1,252	40.0	50.0	33.1
expirationDateTime	281	56.0	61.8	34.7
refillAmount	186	56.5	65.5	39.4
remainingAmount	100	57.0	65.0	39.4
prescribedAmount	40	88.5	89.4	32.7
Whole cohort	164,072	29.0	39.3	32.3

Figure 24: DUR duration by correction field name, Cohort 3.



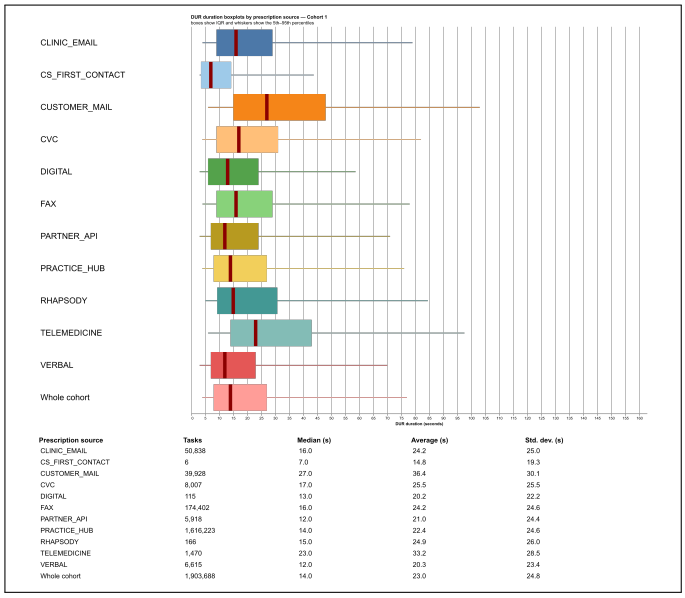
Correction field name	Tasks	Median (s)	Average (s)	Std. dev. (s)
<No correction>	11,338	37.0	46.4	33.0
directions	5,607	43.0	51.9	34.1
notes	769	74.0	78.1	29.3
daysOfSupply	379	63.0	68.9	36.1
prescribedDateTime	181	49.0	61.3	36.9
expirationDateTime	52	74.0	81.8	35.5
refillAmount	49	73.0	72.6	36.8
prescribedAmount	35	79.0	79.2	32.3
remainingAmount	24	72.5	74.5	37.6
Whole cohort	18,434	41.0	50.3	34.2

Figure 25: DUR duration by correction field name, Cohort 4.

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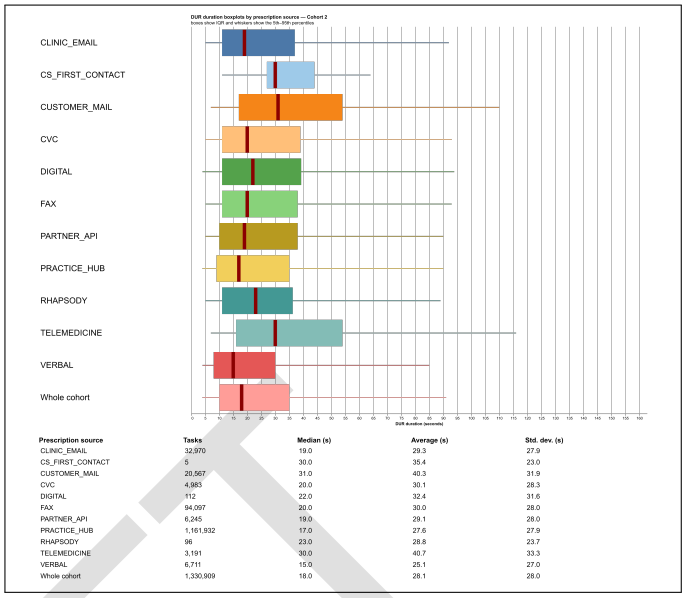
2.9.5 Prescription Source/Approval Channel

Figures 26–29 compare DUR durations by prescription source across Cohorts 1–4. The one thing that stands out is that prescriptions that arrive with customer email take longer time.



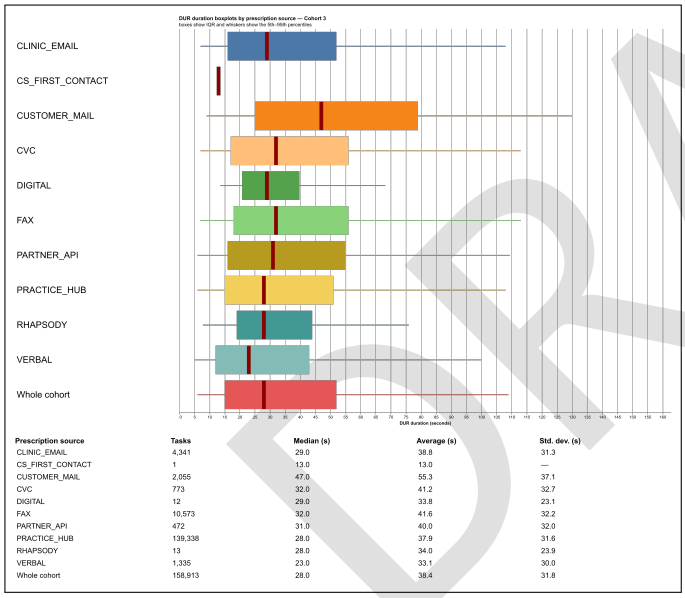
Prescription source	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	50,838	16.0	24.2	25.0
CS_FIRST_CONTACT	6	7.0	14.8	19.3
CUSTOMER_MAIL	39,928	27.0	36.4	30.1
CVC	8,007	17.0	25.5	25.5
DIGITAL	115	13.0	20.2	22.2
FAX	174,402	16.0	24.2	24.6
PARTNER_API	5,918	12.0	21.0	24.4
PRACTICE_HUB	1,616,223	14.0	22.4	24.6
RHAPSODY	166	15.0	24.9	26.0
TELEMEDICINE	1,470	23.0	33.2	28.5
VERBAL	6,615	12.0	20.3	23.4
Whole cohort	1,903,688	14.0	23.0	24.8

Figure 26: DUR duration by prescription source, Cohort 1.



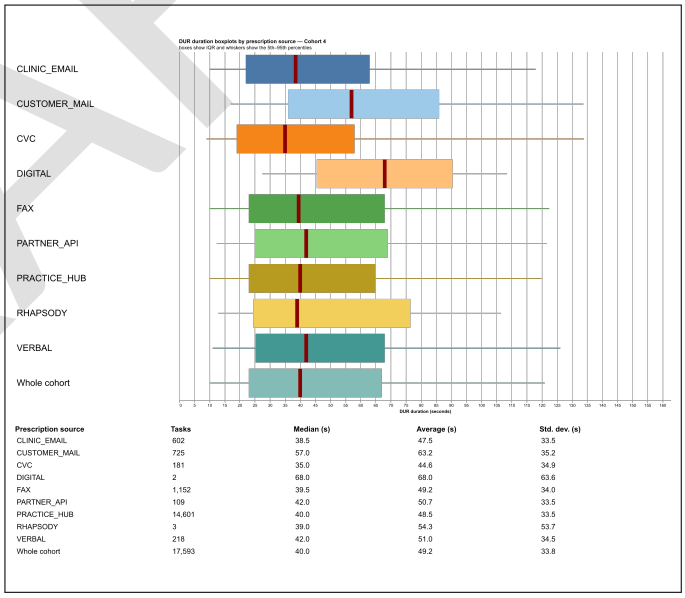
Prescription source	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	32,970	19.0	29.3	27.9
CS_FIRST_CONTACT	5	30.0	35.4	23.0
CUSTOMER_MAIL	20,567	31.0	40.3	31.9
CVC	4,983	20.0	30.1	28.3
DIGITAL	112	22.0	32.4	31.6
FAX	94,097	20.0	30.0	28.0
PARTNER_API	6,245	19.0	29.1	28.0
PRACTICE_HUB	1,161,932	17.0	27.6	27.9
RHAPSODY	96	23.0	28.8	23.7
TELEMEDICINE	3,191	30.0	40.7	33.3
VERBAL	6,711	15.0	25.1	27.0
Whole cohort	1,330,909	18.0	28.1	28.0

Figure 27: DUR duration by prescription source, Cohort 2.



Prescription source	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	4,341	29.0	38.8	31.3
CS_FIRST_CONTACT	1	13.0	13.0	—
CUSTOMER_MAIL	2,055	47.0	55.3	37.1
CVC	773	32.0	41.2	32.7
DIGITAL	12	29.0	33.8	23.1
FAX	10,573	32.0	41.6	32.2
PARTNER_API	472	31.0	40.0	32.0
PRACTICE_HUB	139,338	28.0	37.9	31.6
RHAPSODY	13	28.0	34.0	23.9
VERBAL	1,335	23.0	33.1	30.0
Whole cohort	158,913	28.0	38.4	31.8

Figure 28: DUR duration by prescription source, Cohort 3.

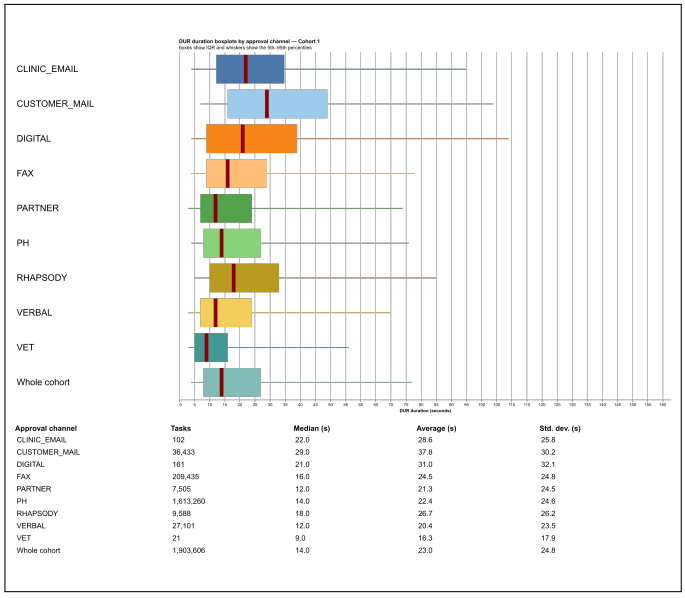


Prescription source	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	602	38.5	47.5	33.5
CUSTOMER_MAIL	725	57.0	63.2	35.2
CVC	181	35.0	44.6	34.9
DIGITAL	2	68.0	68.0	63.6
FAX	1,152	39.5	49.2	34.0
PARTNER_API	109	42.0	50.7	33.5
PRACTICE_HUB	14,601	40.0	48.5	33.5
RHAPSODY	3	39.0	54.3	53.7
VERBAL	218	42.0	51.0	34.5
Whole cohort	17,593	40.0	49.2	33.8

Figure 29: DUR duration by prescription source, Cohort 4.

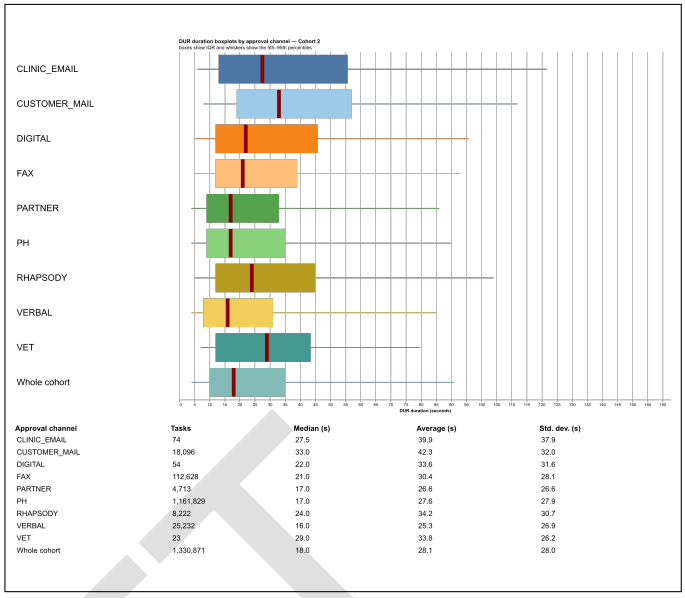
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Figures 30–33 compare DUR durations by approval channel across the same cohorts. Customer email is associated with longer DUR times.



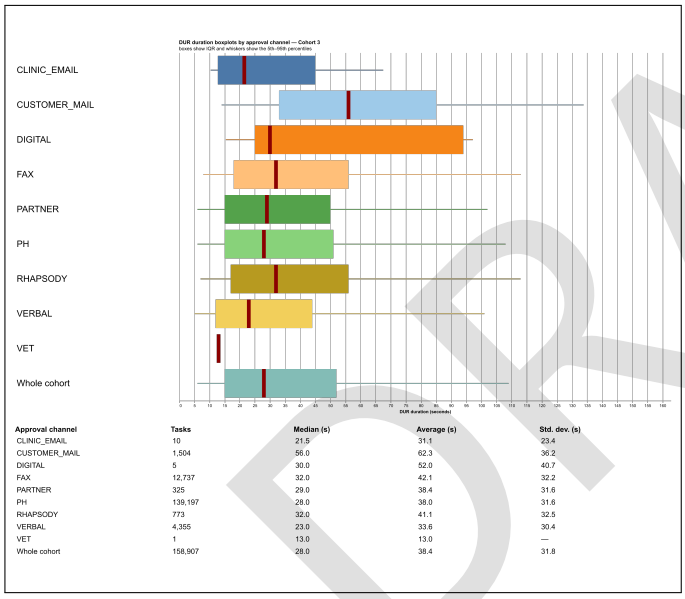
Approval channel	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	102	22.0	28.6	25.8
CUSTOMER_MAIL	36,433	29.0	37.8	30.2
DIGITAL	161	21.0	31.0	32.1
FAX	209,435	16.0	24.5	24.8
PARTNER	7,505	12.0	21.3	24.5
PH	1,613,260	14.0	22.4	24.6
RHAPSODY	9,588	18.0	26.7	26.2
VERBAL	27,101	12.0	20.4	23.5
VET	21	9.0	16.3	17.9
Whole cohort	1,903,606	14.0	23.0	24.8

Figure 30: DUR duration by approval channel, Cohort 1.



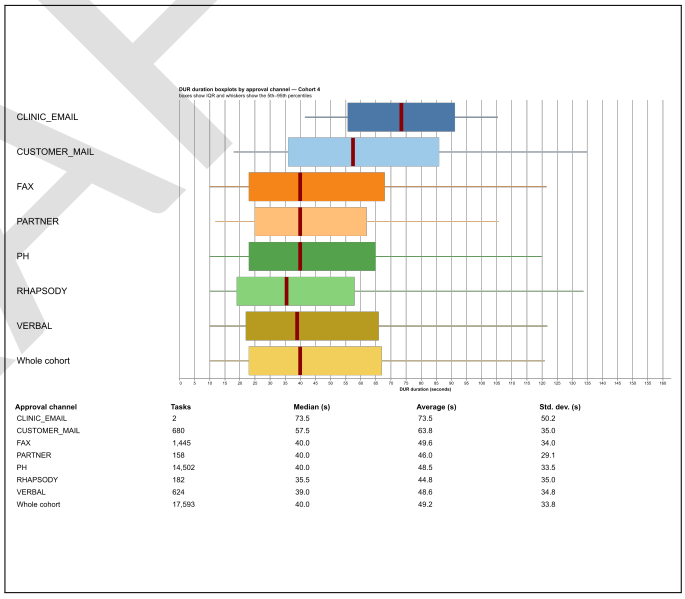
Approval channel	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	74	27.5	39.9	37.9
CUSTOMER_MAIL	18,096	33.0	42.3	32.0
DIGITAL	54	22.0	33.6	31.6
FAX	112,628	21.0	30.4	28.1
PARTNER	4,713	17.0	26.6	26.6
PH	1,161,829	17.0	27.6	27.9
RHAPSODY	8,222	24.0	34.2	30.7
VERBAL	25,232	16.0	25.3	26.9
VET	23	29.0	33.8	26.2
Whole cohort	1,330,871	18.0	28.1	28.0

Figure 31: DUR duration by approval channel, Cohort 2.



Approval channel	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	10	21.5	31.1	23.4
CUSTOMER_MAIL	1,504	56.0	62.3	36.2
DIGITAL	5	30.0	52.0	40.7
FAX	12,737	32.0	42.1	32.2
PARTNER	325	29.0	38.4	31.6
PH	139,197	28.0	38.0	31.6
RHAPSODY	773	32.0	41.1	32.5
VERBAL	4,355	23.0	33.6	30.4
VET	1	13.0	13.0	—
Whole cohort	158,907	28.0	38.4	31.8

Figure 32: DUR duration by approval channel, Cohort 3.



Approval channel	Tasks	Median (s)	Average (s)	Std. dev. (s)
CLINIC_EMAIL	2	73.5	73.5	50.2
CUSTOMER_MAIL	680	57.5	63.8	35.0
FAX	1,445	40.0	49.6	34.0
PARTNER	158	40.0	46.0	29.1
PH	14,502	40.0	48.5	33.5
RHAPSODY	182	35.5	44.8	35.0
VERBAL	624	39.0	48.6	34.8
Whole cohort	17,593	40.0	49.2	33.8

Figure 33: DUR duration by approval channel, Cohort 4.

3 References

[1] *Would Grouping Similar Work Make Pharmacists Faster? What the data says*. Internal analysis. https://chewyinc.atlassian.net/wiki/x/hYRyRQE.

[2] JD Wright. Cohort definitions and rationale. Internal conversation, undated; reproduced in the Appendix.

[3] Wikipedia contributors. *Confounding*. https://en.wikipedia.org/wiki/Confounding.

[4] *RXP Pharmacist Task Grouping Analysis*. GitHub repository. https://github.com/myunt1-chwy/rxp-pharmacist-task-grouping-analysis.

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